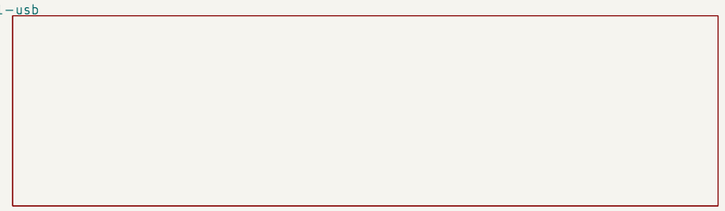
OPENKEYS R0.5 – ONE USB-C + NFC / BLUE 16x2 LCD
Device PIN mandatory / TEN DECIMAL KEYS / NO FINGERPRINT
PCB candidate; qualification required before secret provisioning

01-usb



File: 01-usb.kicad_sch

02-power



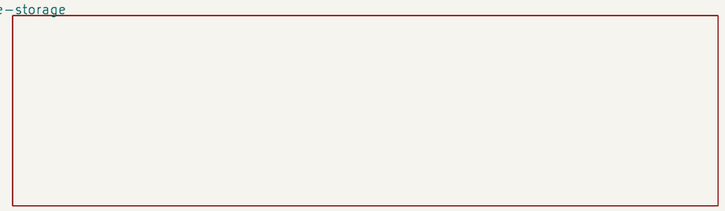
File: 02-power.kicad_sch

03-mcu



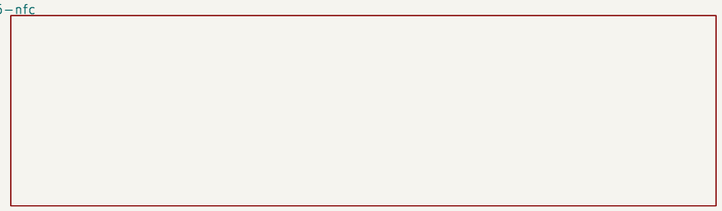
File: 03-mcu.kicad_sch

04-se-storage



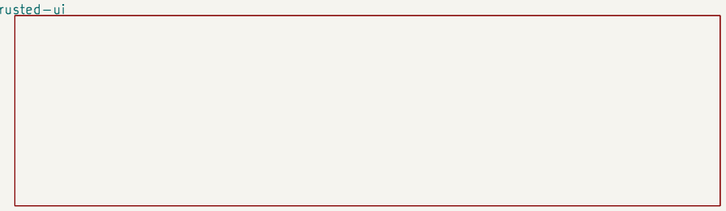
File: 04-se-storage.kicad_sch

05-nfc



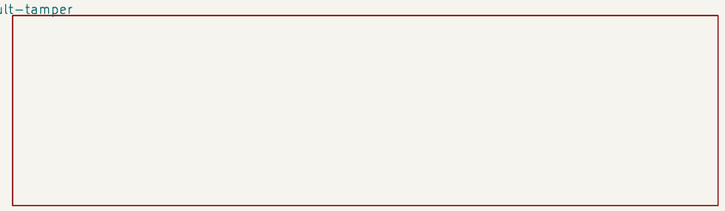
File: 05-nfc.kicad_sch

06-trusted-ui



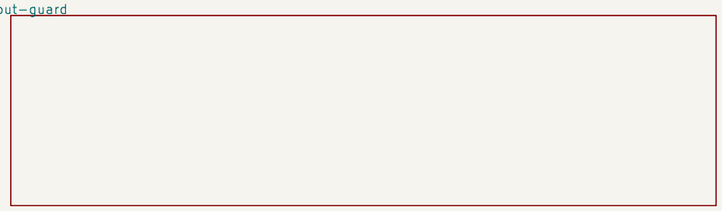
File: 06-trusted-ui.kicad_sch

07-fault-tamper



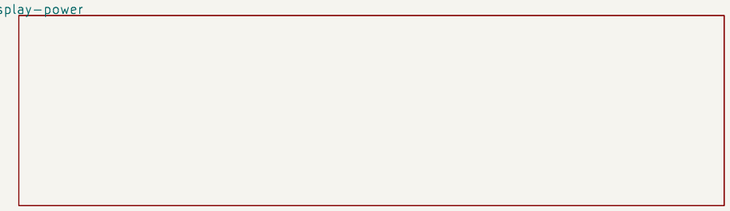
File: 07-fault-tamper.kicad_sch

08-input-guard



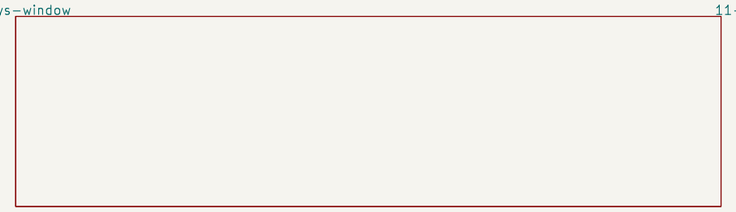
File: 08-input-guard.kicad_sch

09-display-power



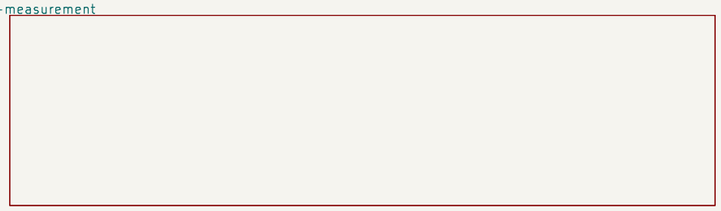
File: 09-display-power.kicad_sch

10-sys-window



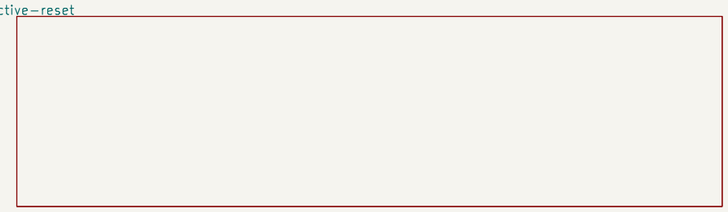
File: 10-sys-window.kicad_sch

11-power-measurement



File: 11-power-measurement.kicad_sch

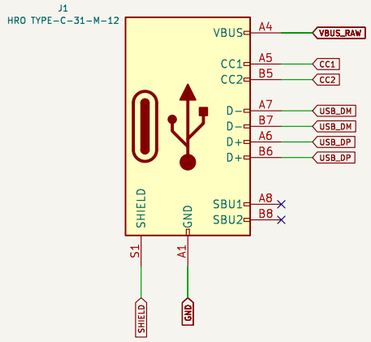
12-active-reset

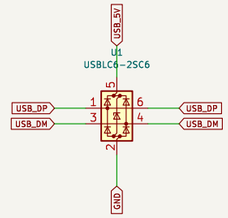


File: 12-active-reset.kicad_sch

Jay-Void-BA
Sheet: /
File: OpenKeys.kicad_sch
Title: OpenKeys R0.5 – Engineering review
Size: A3 | Date: 2026-11-13 | **Rev: 0.5**
KiCad E.D.A. 9.0.2 | Id: 1/13

One USB-C receptacle — USB 2.0 FS data and 5 V charging

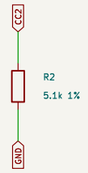


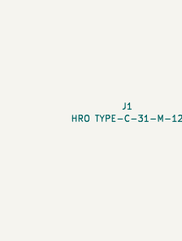


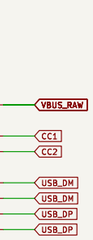




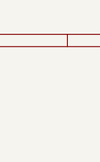


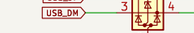


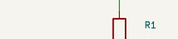














5.1k Rd on each CC pin: sink only. No PD negotiation, no 9 V request.
Apple 20 W adapter uses 5 V through a standards-compliant USB-C cable.
100mA hardware input limit: 5 V x 0.1 A = 0.50 W nominal input budget.
Input guard U14 is independent of MCU: see sheet08. Analog timing/thermal protection not yet qualified.

ENGINEERING BOARD CANDIDATE — qualify before use

Jay-Void-BA

Sheet: /01-usb/
File: 01-usb.kicad_sch

Title: OpenKeys R0.5 — 01-usb

Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 2/13

Low-rate charging, protected 500 mAh pack and 3.3 V supply

#FLG3
PWR_FLAG

VBUS_FUSED

VBUS_RAW

F1
250mA hold – qualify I2t

VBUS_FUSED

U2
BQ25185DLHR

SYS	1	SYS	
BAT	2	BAT	
CHG_STAT2_N	3	STAT2	
CHARGE_DISABLE	4	CE_N	
GND	5	GND	
BAT_NTC	6	TS_MR	
		ILIM_VSET	7
		ISET	8
		STAT1	9
		IN	10
		EP	11

ILIM_VSET
CHG_ILIM_VSET

ISET
CHG_ISET

STAT1
CHG_STAT1_N

IN
USB_5V

EP
GND

J2
Adafruit 1578 protected 1S 500mAh

BAT
GND

CHG_ILIM_VSET

R6
24k 1%

GND

CHG_ISET

R7
3.00k 1%

GND

SYS

R8
100k

CHARGE_DISABLE

CHARGE_DISABLE

CHARGE_ENABLE

1

Q2
2N7002

2

3

GND

CHARGE_ENABLE

R9
100k

GND

J6
103AT-2 external battery thermistor

BAT_NTC
GND

1
2

SW_L1

L1
1.5uH Isat>=1.5A

SW_L2

USB_5V

C2
1uF 10V

GND

BAT

C3
10uF 6.3V

GND

SYS

C4
10uF 10V

GND

SYS

R10
0R

REG_VINA

#FLG4
PWR_FLAG

REG_VINA

U3
TPS63031DSK (3.3 V)

SYS	5	VIN	
REG_VINA	8	VINA	
REG_EN	6	EN	
SYS	7	PS/SYNC	
SW_L1	4	L1	
SW_L2	2	L2	
GND	9	GND	
GND	3	PGND	
		VOUT	1
		FB	10

VOUT
+3V3

FB
+3V3

REG_EN

2

SW1

1
3

SYS

OFF / ON

GND

SYS

C5
10uF 10V

GND

REG_VINA

C6
100nF

GND

+3V3

C7
10uF 6.3V

GND

+3V3

C8
10uF 6.3V

GND

Fixed100mA input; charge target100mA=0.2C for500mAh.
Input DPM gives system load priority, so cell charge can be slower.
CE defaults disabled. Public charging service never unlocks secrets.
Suspend: stop charge/backlight; qualify total VBUS current <=2.5mA.
Real pack PCM + attached thermistor required. No fast charge.

ENGINEERING BOARD CANDIDATE – qualify before use

Jay-Void-BA

Sheet: /02-power/
File: 02-power.kicad_sch

Title: OpenKeys R0.5 – 02-power

Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 3/13

STM32H573RIT6 — STiRoT and TrustZone protected I/O

C9
100nF

C10
100nF

C11
100nF

C12
100nF

C13
100nF

C14
4.7uF

C15
2.2uF low ESR

C16
2.2uF low ESR

C17
100nF

C18
1uF

U4
STM32H573RIT6

FB1
600R@100MHz

#FLG6
PWR_FLAG

R11
10k

R12
10k

C19
10nF — qualify reset timing

C20
9pF C0G — tune

C21
9pF C0G — tune

Y1
ABS07-32768KHZ-6-T

J3
Internal SWD — DNP production

STiRoT must verify protected firmware. TrustZone/GTZC must deny nonsecure access to SE, LCD, ten keys, timers, DMA and key memory.
No raw cryptography or arbitrary display commands at secure gateway.
SE_EN is a fault interlock, not proof of a correct PIN.
Separate2.2uF on VCAP1/2; no external load on VCAP.

ENGINEERING BOARD CANDIDATE — qualify before use

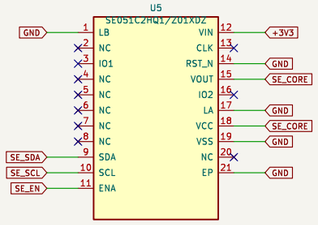
Jay-Void-BA

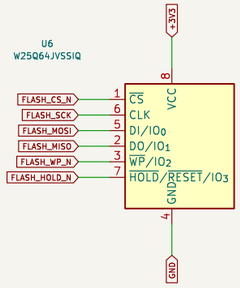
Sheet: /03-mcu/
File: 03-mcu.kicad_sch

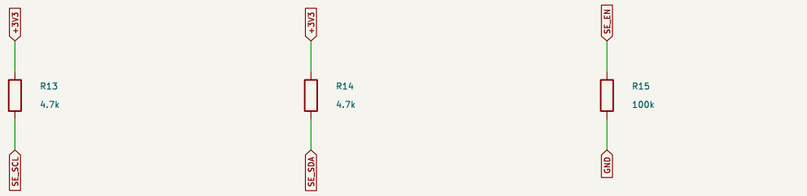
Title: OpenKeys R0.5 — 03-mcu

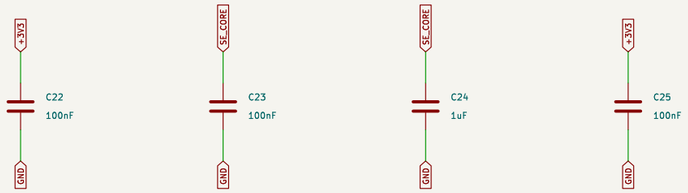
Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 4/13

Secure element, protected keys and encrypted records









Private operations: valid local six-digit PIN grant + per-request confirmation.
No biometric path. SE-native PIN ACLs and durable retries must be qualified; Root owns keypad.
Precharge attempts durably; uncertain writes deny. AEAD does not alone prevent rollback.
SE certification does not certify MCU, PCB, PIN execution path or complete product.

ENGINEERING BOARD CANDIDATE — qualify before use

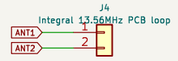
Jay-Void-BA

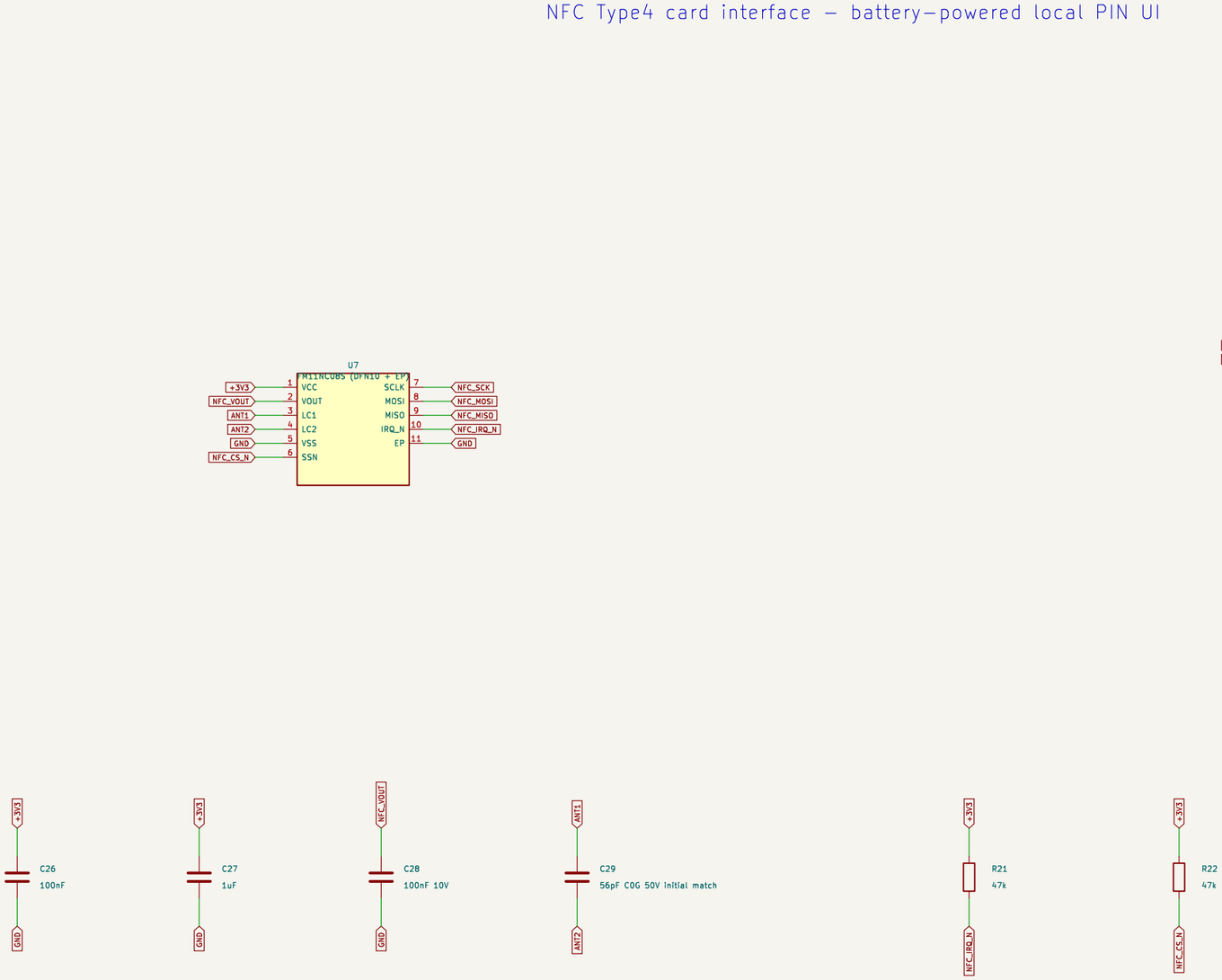
Sheet: /04-se-storage/
File: 04-se-storage.kicad_sch

Title: OpenKeys R0.5 — 04-se-storage

Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 5/13

NFC Type4 card interface — battery-powered local PIN UI





NFC VOUT isolated from SYS/BAT/+3V3. No assumed RF-powered full device.
LCD/PIN/CTAP NFC require charged battery and device ON.
RF region: tuned coil only, no cell, metal shield or ground planes.
Mobile management needs authenticated APDUs and native OS integration.
Phone biometrics or a tap never replace device PIN verification.

ENGINEERING BOARD CANDIDATE — qualify before use

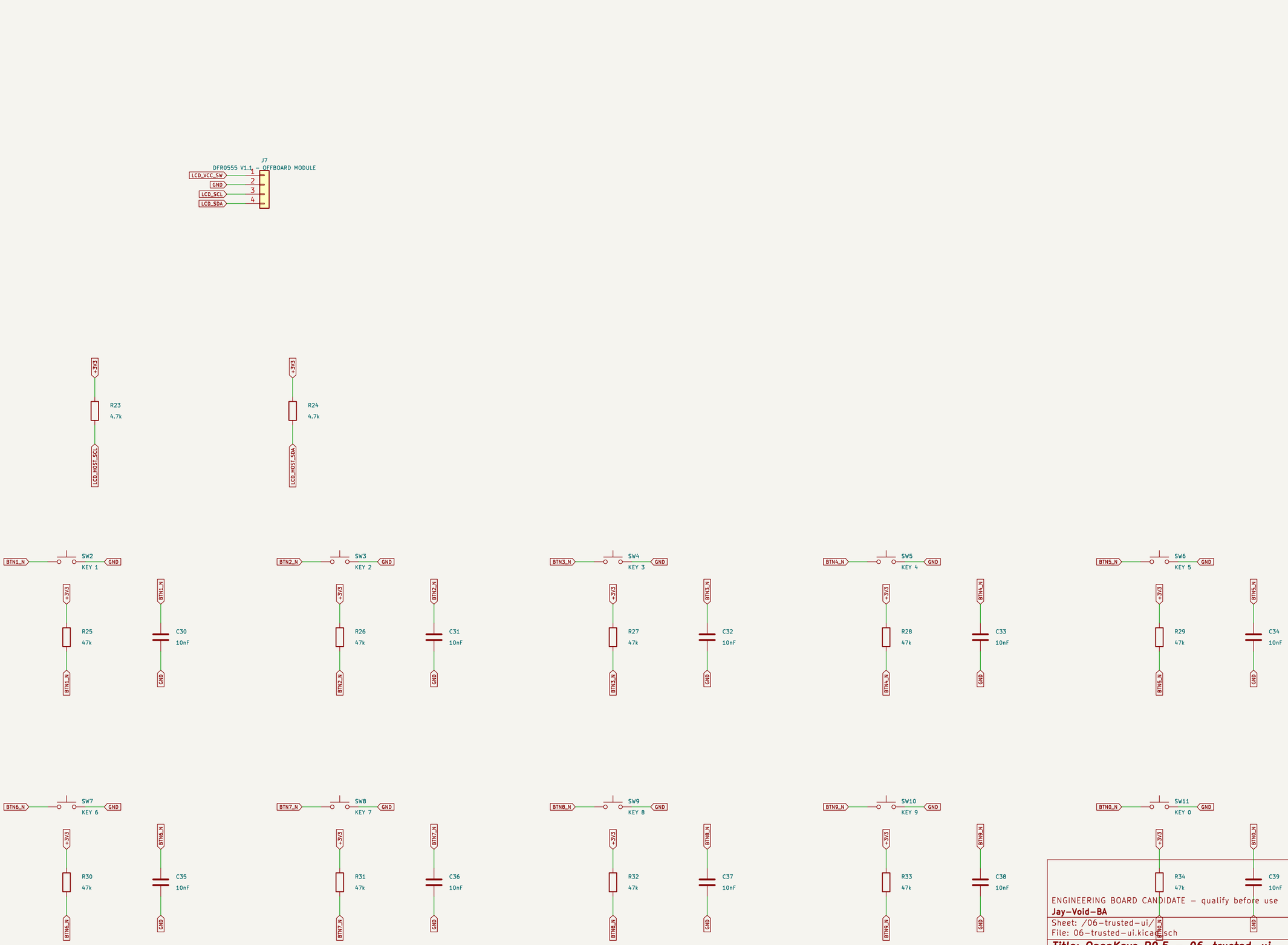
Jay-Void-BA

Sheet: /05-nfc/
File: 05-nfc.kicad_sch

Title: OpenKeys R0.5 — 05-nfc

Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 6/13

DFR0555 blue 16x2 LCD and ten independent PIN keys



ENGLISH 16x2, blue background / white text. No fingerprint parts.
PIN:6 decimal digits, masked. Hold9=erase; hold0=cancel.
Review:2=previous,8=next,5=confirm,0=cancel. New release required.
Private operations require a valid paired3Os local PIN grant and exact review.
No PIN from USB/NFC/desktop. No factory/default unlock PIN.

ENGINEERING BOARD CANDIDATE – qualify before use

Jay-Void-BA

Sheet: /06-trusted-ui/
File: 06-trusted-ui.kicad_sch

Title: *OpenKeys R0.5 – 06-trusted-ui*

Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 7/13

Independent backup cell, active case loop, reset and SE interlock

BT1
Panasonic CR-2032/HFN
BACKUP_CELL
GND

BACKUP_CELL
R61
10k 1%
BACKUP_LIMITED

U10
TPS3702CX33DDCR
SENSE
SET
VDD
GND
OV
UV
+3V3
+3V3
NRST
NRST
SYS
BACKUP_LIMITED
GND

U13
LM66100DCKR
VIN
GND
CE
NC
ST_UNUSED_GND
VOUT
BACKUP_LIMITED
GND
VBAT_BACKUP
GND
VBAT_BACKUP

BACKUP_LIMITED
C40
100nF X7R 16V
GND

VBAT_BACKUP
C41
4.7uF 10V X7R
GND

VBAT_BACKUP
C56
100nF X7R 16V
GND

+3V3
SW12
LID RESET contact
CASE_CLOSED

CASE_CLOSED
R35
100k
GND

#FLG_BACKUP0101
PWR_FLAG
BACKUP_LIMITED

U11
SN74LVC1G07DCKR
GND
VCC
NC
+3V3
CASE_CLOSED
NRST
GND

U12
SN74LVC1G08DCKR
GND
VCC
NRST
SE_ENABLE_REQUEST
SE_EN
+3V3
GND

SE_ENABLE_REQUEST
R36
100k
GND

SYS
C42
100nF 10V
GND

+3V3
C43
100nF
GND

+3V3
C44
100nF
GND

CASE_TAMP_RETURN
C68
100pF C0G 50V 5%
GND

TAMP_DRIVE_RAW
R63
1k 1%
TAMP_DRIVE

TAMP_DRIVE
SW13
ACTIVE LID 3
TAMP_LID_MID1

TAMP_LID_MID1
SW14
ACTIVE LID 4
TAMP_LID_MID2

TAMP_LID_MID2
SW15
ACTIVE LID 5
CASE_TAMP_RETURN

CASE_TAMP_RETURN
R62
470k 1%
GND

PA1=TAMP_OUT4 -> three lid contacts -> PC13=TAMP_IN1; hardware active shared output4.
Separate SW12 asynchronously resets MCU/LCD and disables SE while main power is present.
PC1 is NOT tied to NRST: it is not a VBAT-capable input and a reset is not a lid event.
Backup register anchor + active engine continuity must be checked; loss/flag => quarantine, never factory blank.
No automatic flag clear/reseed on normal boot. No proof against bridging, probing, SCA or all glitches.
Qualify backup life, reverse current, RF, reset latency, mechanics and injection before release.

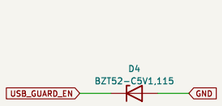
ENGINEERING BOARD CANDIDATE – qualify before use
Jay-Void-BA

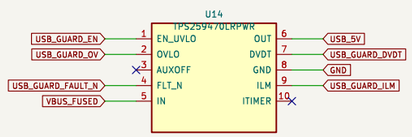
Sheet: /07-fault-tamper/
File: 07-fault-tamper.kicad_sch

Title: OpenKeys R0.5 – 07-fault-tamper

Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 8/13

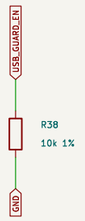
Independent VBUS overvoltage cutoff, reverse blocking and limited inrush

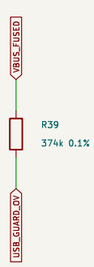


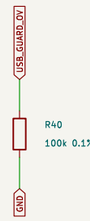


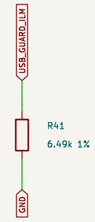


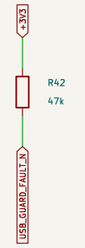


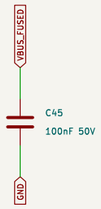


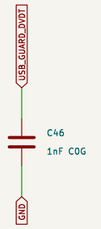












OVLO nominal1.2V x(1+374k/100k)=5.688V. Component/leakage corners in protection-calculations.json.
UVLO divider22.1k/10k: about3.85V nominal. EN is separately clamped, never raw VBUS.
6.49k ILM: about0.51A protection threshold; BQ charger still sets100mA input budget.
1nF DVDT: about2V/ms, about2.5ms ramp to5V; measure actual inrush with complete load.
F1 -> input TVS -> eFuse -> output TVS/ESD reference/charger/sense.
Independent cutoff is a circuit provision; response overshoot, CC/data faults and thermal/surge tests remain BLOCKED.

ENGINEERING BOARD CANDIDATE - qualify before use

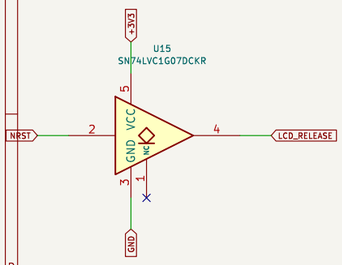
Jay-Void-BA

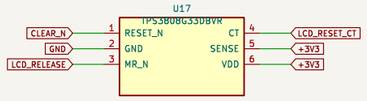
Sheet: /08-input-guard/
File: 08-input-guard.kicad_sch

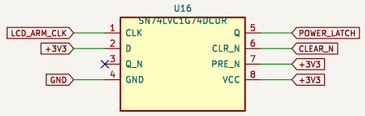
Title: OpenKeys R0.5 - 08-input-guard

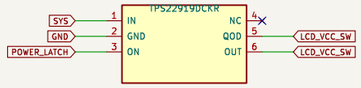
Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 9/13

Latched LCD power removal; isolated bus; no resetless retained text

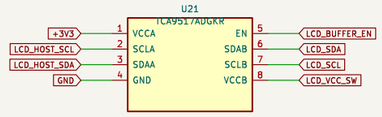


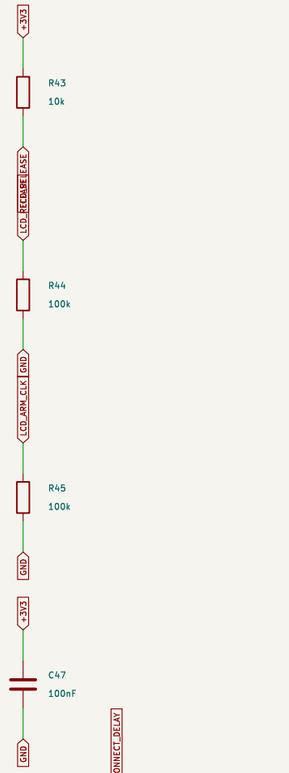


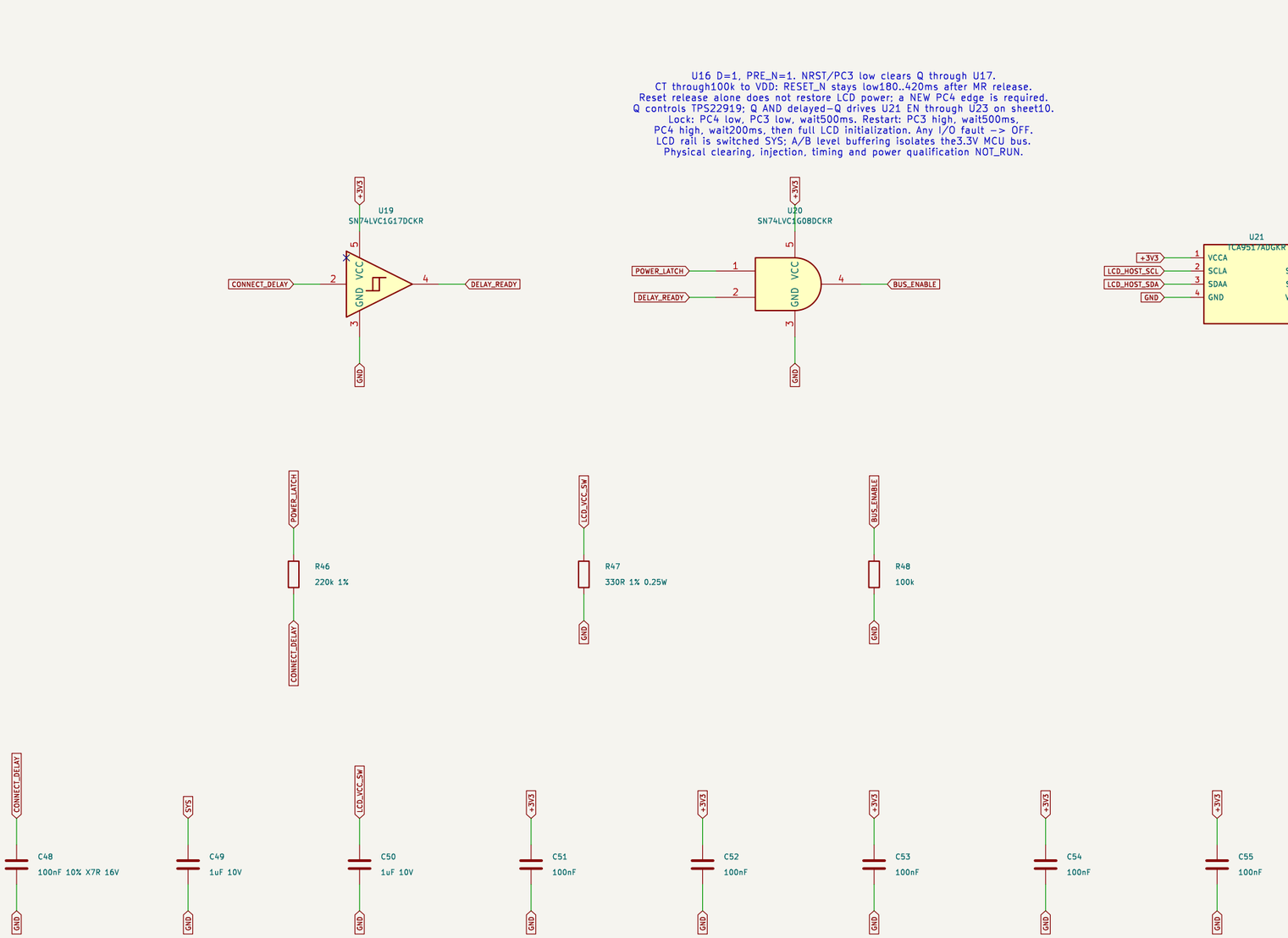




U16 D=1, PRE_N=1. NRST/PC3 low clears Q through U17.
CT through100k to VDD: RESET_N stays low180..420ms after MR release.
Reset release alone does not restore LCD power; a NEW PC4 edge is required.
Q controls TPS22919; Q AND delayed-Q drives U21 EN through U23 on sheet10.
Lock: PC4 low, PC3 low, wait500ms. Restart: PC3 high, wait500ms,
PC4 high, wait200ms, then full LCD initialization. Any I/O fault -> OFF.
LCD rail is switched SYS; A/B level buffering isolates the3.3V MCU bus.
Physical clearing, injection, timing and power qualification NOT_RUN.

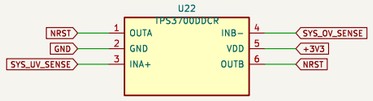


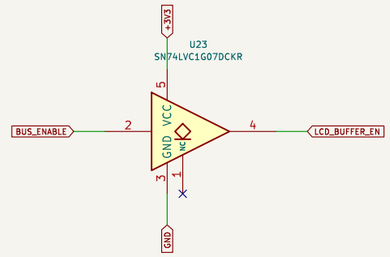




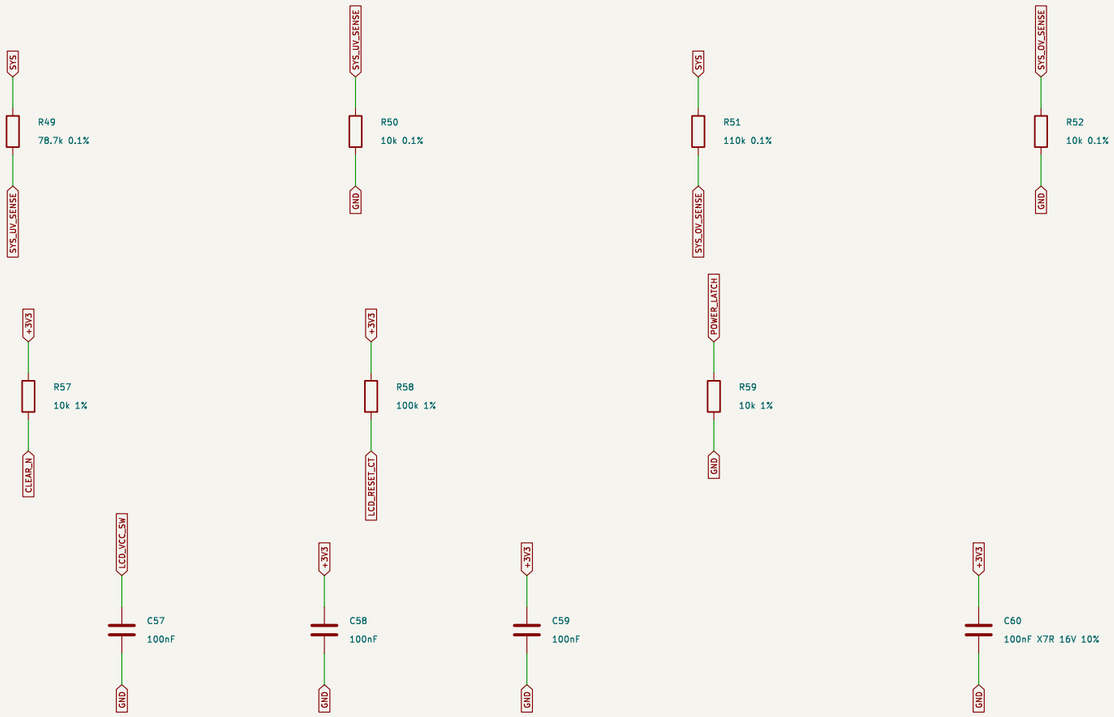
ENGINEERING BOARD CANDIDATE — qualify before use
Jay-Void-BA
Sheet: /09-display-power/
File: 09-display-power.kicad_sch
Title: OpenKeys R0.5 — 09-display-power
Size: A3 | Date: 2026-11-13 | **Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2 | Id: 10/13

SYS voltage window and LCD level–buffer fail–closed enable





TPS3700 reference: positive396..404mV; negative387..400mV.
Include0.1percent divider errors and +/–25nA input bias.
Static SYS UV fall min3.4246V; OV rise max4.8596V.
With ASSUMED switch+harness drop<=50mV: LCD>=3.3746V.
Main3.3V fixed PWM:3.267..3.333V; no ripple/transient guarantee.
TCA sideA=main3.3V; sideB=switched SYS. Off I2C must stay high–Z.
EN high follows switched VCCB through U23, R53/R54.
LCD total C<=100uF and aggregate off injection<=20uA MUST be measured.
Hardware hold180ms minus ASSUMED turn–off<=5ms permits discharge.
Services stay locked during startup, undervoltage and all faults.
Fast pulses, backfeed, input budget, temperature and RF remain unqualified.



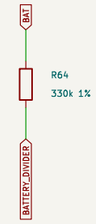
ENGINEERING BOARD CANDIDATE – qualify before use
Jay–Void–BA

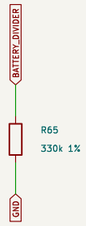
Sheet: /10–sys–window/
File: 10–sys–window.kicad_sch

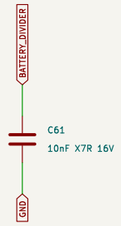
Title: OpenKeys R0.5 – 10–sys–window

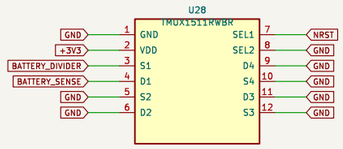
Size: A3	Date: 2026–11–13	**Rev: 0.5–board–review**
KiCad E.D.A. 9.0.2		Id: 11/13

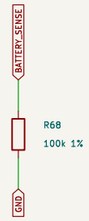
Protected main-cell voltage measurement; no invented battery percentage

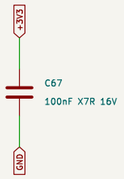












PA0=ADC1_INP0 analog only. Read VREFINT/calibration; use long sample time and settled bounded samples.
330k/330k divider +100k output bias: nominal Vadc/Vbat=100/530 when U28 is on.
Reservoir is upstream of U28. NRST low disconnects ADC; powered-off path must not backfeed main rail.
Voltage/charging is not a fuel gauge; no precise percentage without qualified pack curve.
No raw ADC/high-rate power telemetry to host; no sampling/return during private cryptography.
Status never renews PIN leases. Internal VBAT charger must stay disabled. Power ramps/leakage NOT tested.

ENGINEERING BOARD CANDIDATE — qualify before use

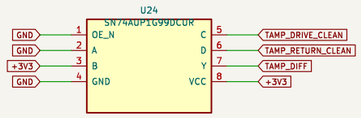
Jay-Void-BA

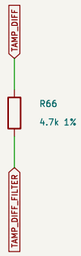
Sheet: /11-power-measurement/
File: 11-power-measurement.kicad_sch

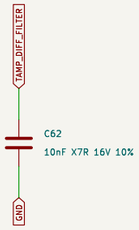
Title: OpenKeys R0.5 — 11-power-measurement

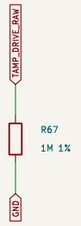
Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 12/13

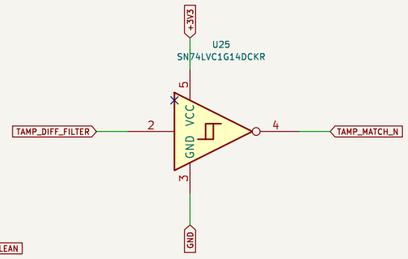
Independent active-loop mismatch reset; no software arm or override

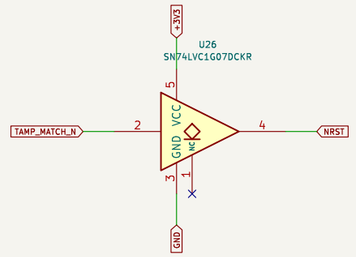


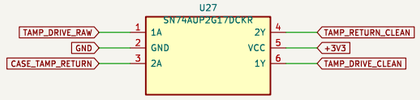




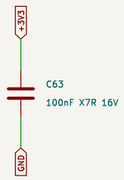


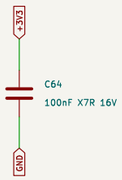


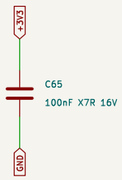


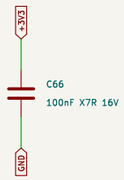












Closed loop: C=D => no reset. Open/stuck/grounded loop: mismatch pulses feed RC and hard reset.
U24 OE and function straps are fixed copper, not GPIO. No host bypass or software mask.
U27 Ioff limits feed into unpowered main logic; leakage is not zero. No maximum input slew at U27/U25.
Active TAMP hardware, backup anchor and fresh secure readback remain mandatory; XOR alone is insufficient.
Probe bridging, matched waveform injection and short glitches are residual risks requiring lab tests.
Powered/unpowered transitions, input thresholds, reset propagation and backup life NOT qualified.

ENGINEERING BOARD CANDIDATE – qualify before use

Jay-Void-BA

Sheet: /12-active-reset/
File: 12-active-reset.kicad_sch

Title: OpenKeys R0.5 – 12-active-reset

Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 13/13